

TRACE: what a memory keeps

Associative memory and fast weights | One claim. A visible failure. A reproducible repair.

The claim and the pressure

A fixed-shape associative state can accept additional writes without adding a slot per input, but overlapping cues and competing values can corrupt recall. TRACE makes this claim inspectable for data scientists familiar with vectors and dot products. Conventional attention keeps explicit key/value histories. Compressing associations into a recurrent matrix makes state size independent of sequence length, while introducing a capacity and interference trade-off. This is a statement about a memory mechanism, not a promise of unlimited lossless context.

A mechanism the learner can test

Our state M has nine key features and eight value features: 72 float64 cells. A Hebbian update adds the outer product of key k and value v , after multiplying the previous state by retention λ . A query reads M by a dot product. Constructed keys share a component controlled by overlap ρ ; increasing ρ makes distinct addresses more similar. The value readout passes through a trained softmax decoder. The learner edits pairs and controls, sees every matrix update, and compares prediction with the latest written target.

A falsifiable sixty-second experiment

Write Atlas-Amber twice, then Atlas-Violet. Additive memory retains competing values and recalls Amber even though the latest target is Violet. Switch to a delta update, which writes the residual between the target value and the current readout: recall becomes Violet. The state remains 72 cells. With overlapping keys, such a correction can also disturb other memories. The experiment demonstrates both a useful update rule and the limit of treating all stored associations as independent.

What connects to BDH

Dragon Hatchling interprets working memory through transient synaptic changes. BDH-GPU combines neuron-aligned linear attention with sparse positive activations and learned ReLU-low-rank transformations [1]. TRACE isolates the associative write/read motif; it omits those full architectural features, uses constructed cues and signed value codes, and does not reproduce a BDH language model. Pathway's derivation explains why the broader representation and computational structure matter, rather than merely replacing softmax attention in a small vector space [5].

BDH-CQ and the research landscape

BDH-CQ combines context-updated recurrent memory with latent iterative reasoning [2]. Its paper reports 29.5% pass@2 on public ARC-AGI-1 for 150M parameters, including a black-box audit by Bielik and NYU co-authors without weights access. This is benchmark evidence; TRACE has not replicated the system. Its exact updates remain proprietary. Gated DeltaNet combines gating with delta updates [3]; Titans learns neural memory at test time [4]. They explore different control mechanisms, with full-model evidence on their own evaluation suites. TRACE does not establish an accuracy or cost ranking between them.

Evidence, maturity and limits

TRACE trains 72 decoder parameters on 6,000 generated noisy-symbol samples and tests on 1,500 independent draws, seed 42. The local run classifies 1,499 test samples correctly. This easy synthetic decoding score does not measure long-context reasoning or BDH capability. Live retrieval tests separately cover clean recall, collision, interference and decay. Full-model advantages remain conditional on published experiments; this prototype has no learner-study evidence yet. It is mature enough for a reproducible mechanism demo, not for claims of general AI memory.

What the artifact includes

The public artifact executes the same Python model through Pyodide in a browser worker; its notebook is device-local. The Docker edition uses FastAPI and SQLite with account-based persistence. Playback shows computed states, with inputs and model hash available for inspection. The 576-byte count excludes weights, traces and application overhead. Sequences are capped at 32 writes; retention acts per write, not wall-clock time. Saving a run is application persistence, not durable model learning.

Primary sources

- [1] Kosowski et al. (2025). *The Dragon Hatchling*. [arXiv:2509.26507](#).
- [2] Engdahl et al. (2026). *BDH-CQ*. [arXiv:2608.09888](#).
- [3] Yang et al. (2024 / ICLR 2025). *Gated Delta Networks*. [arXiv:2412.06464](#).
- [4] Behrouz et al. (2025). *Titans*. [arXiv:2501.00663](#).
- [5] Pathway (2026). *From attention to synapses: deriving BDH*.
[Open the public artifact](#) | [Source: Yeshwanthv123/Dataforge2026](#)